JFNK, marching, and JFNK again from one molecular LHS 1140 b state (He/H 0.1, $s = 2.53$)

Summary

Verdict. Neither JFNK attempt produced a stationary state, and marching the first JFNK last iterate did not bring it back to the marching state within the 31 311 s (0.106 flow-crossing times) that were marched. Some differences decreased (the largest density difference from 0.226 to 0.199 dex), but an overall return was not observed. The two attempts show similar late-pass diagnostics, especially in passes 36–48. Their uncertified last iterates have opposite helium-rich and helium-poor distributions and differ strongly from each other and from the campaign state.

- **JFNK-1** (from chunk 61) and **JFNK-2** (from the end of the remarch): 60 outer passes each, `info=1`, exit status 3, certified=F. The reported hydrodynamic rows of the retained wind remain small (mass $\leq 6.3 \times 10^{-8}$, momentum $\leq 3 \times 10^{-11}$, energy $\leq 1.6 \times 10^{-8}$); four hydrodynamic solves fail and restore an earlier converged wind (Section 5.). Meanwhile the worst carrier row measured after that solve stays at 0.42–1.0 against 10^{-5} from pass 9 (JFNK-1) and pass 2 (JFNK-2) on.

- **A (carriers).** When the JFNK-1 iterate is marched, H_2^+, H_3^+ and HeH^+ at 1.2–1.5 R_p fall by 0.40–1.03 dex by the first frame (1564.5 s) and He^+, He(2^3S) rise by 0.16–0.36 dex at 1.27–2 R_p; H^+ and e^- in the layer (1.03–1.5 R_p) then change by 0.3–0.75 dex, together with an inflow that develops at 1.03–1.13 R_p. At selected cells these later changes reach half of their measured log-density change after 6000–8500 s. The carriers evolve in the displaced fluid and helium distribution; they have not returned to the chunk-61 values, and some still change during the last chunk.

- **B (fluid).** Against the campaign at equal elapsed time, the largest density difference goes from 0.226 to 0.199 dex (rms 0.082 to 0.083 dex), the temperature difference from 0.075 to 0.071 dex, and the velocity measure grows from 0.09 to 0.42: some differences decrease, but no overall return is observed. A persistent difference is the helium distribution of the JFNK-1 uncertified last iterate, with local He/H up to 13 times the campaign value, in a layer whose advection times exceed 10^6 s.

- **C, D.** JFNK-2 moves its start by up to 0.41 dex in ρ and removes helium from the wind (He/H 1.5×10^{-11} at 10 R_p), the opposite of JFNK-1 (0.045); its $4\pi\rho v r^2$ is 1.87×10^8 g s^{-1} against 7.08×10^7 (JFNK-1) and 7.28×10^7 (chunk 61). The two adjacent saved pass states of each attempt have opposite helium-poor and helium-rich wind distributions. Similar scalar pass diagnostics do not establish repetition of the full states; the displacement is not reproducible as a state.

- **E.** The composition update at fixed wind can reduce the worst carrier row to 5×10^{-5} (still above 10^{-5}), but the next hydrodynamic solve, with transported fractions retained and the usual thermochemical and boundary refreshes applied, raises it to 0.55–1.0 in the listed passes; the march holds them at 10^{-4}–2×10^{-2} (H_2 up to 0.15). The records appear consistent with a partitioned iteration that does not contract for this molecular case; they do not establish the cause term by term.

Both final JFNK outputs are uncertified last iterates (certified=F). The saved states of earlier passes are intermediate iterates; none is presented as a stationary solution.

1. Setup

Three calculations start, one after the other, from the stored state of campaign chunk 61 of the He/H 0.1, $s = 2.53$ marching phase ($t_{\mathrm{phys}} = 552823.11$ s; flow-crossing time $\tau_{\mathrm{cross}} = 2.943 \times 10^5$ s in the phase progress record and in our own evaluation of the same state). The undisturbed march continues from the same state as campaign chunks 62–66, which serve as the reference at equal elapsed time. Background on the two procedures and on earlier molecular stationary failures is in `stationary_vs_marching_lhs1140b.tex`; it is not repeated here.

1. **JFNK-1**: the existing stationary route (`Restart intent: stationary`) from chunk 61. Each outer pass solves the hydrodynamic rows with the composition held, then relaxes the elements (He/H partition) and the transported carriers at the solved wind (`carrier_steady_residual`, `diffusive_photochemistry.f90` 4288–4304). 60 passes, `info=1`, exit status 3, certified=F. Its written state is the last iterate after the hydrodynamic solve of pass 60 and its thermochemical and boundary refreshes (no element or carrier update follows the last pass).[R1]
2. **Remarch**: the JFNK-1 last iterate marched four campaign chunks (50 001 steps each, CFL 0.15, WENO3, HLLC, `Run mode: phys`), ending at $t = 31311.39$ s. All four chunks were admitted.[RJ]
3. **JFNK-2**: the JFNK-1 settings from the end of remarch chunk 4. 60 passes, `info=1`, exit status 3, certified=F.[R2]

Table 1: Conditions of the stages. All use binary `EXHALE_frozen.x`, sha256 `7e17a6bc0b22f4f9...` (the phase binary; the four copies have equal hashes), 12 OpenMP threads, and the phase side files (equal hashes in the staging records).

	JFNK-1	remarch chunks 1–4	JFNK-2
starts from	chunk 61 triple, byte copy	JFNK-1 last iterate, byte copy	remarch chunk 4 end, byte copy
input vs phase	`Run mode: init`, `Restart intent: stationary`, `Solver: Newton`, `Resid tol: 4.0e-6`	identical	identical to JFNK-1
environment	phase + 5 `EXHALE_*` iteration-control variables	phase + `OMP_NUM_THREADS`	identical to JFNK-1
host	lart3 (Gold 6254)	lart3	lart3
wall	19 935 s (limit 21 600 s)	4281, 4218, 4192, 4198 s	16 158 s
clock	no physical time	restarted at $t = 0$	no physical time

The five variables are `EXHALE_PTC_SER=2`, `EXHALE_JFNK_MAXIT=400`, `EXHALE_DIFF_OMEGA=1.0`, `EXHALE_PTC_DTAU0=1.0`, `EXHALE_OUTER_PASSES=60`.[PLAN 3] The reference campaign chunks ran on lart1 (Gold 6154) with the same binary and input; no bitwise statement across hosts is made.

Caveats on what a remarch frame is. (i) The JFNK output is a `mode=init` file; a phys run takes it as the start of physical integration with $t = 0$ (console of remarch chunk 1, lines 61–69). Remarch time is therefore elapsed time, and remarch chunk k is compared with the campaign state nearest in elapsed time (offset at most 800 s; the campaign drift over 800 s is below 1.2×10^{-4} dex in ρ). (ii) On load the code re-equilibrates the composition at fixed ρ, v, p and recomputes T before the first step (`equilibrate_loaded_composition`). The only number it prints is the particle-plus-electron count it removed, 6.1×10^{-16} (relative), against 1.4×10^{-11} in campaign chunk 62. No state is written between the load and the first frame at step 10 001 ($t = 1564.5$ s), so the effect of the re-equilibration on the individual carriers cannot be separated from the first 1564.5 s of integration. (iii) Remarch chunk 1 refused 106 of 55 318 outer step attempts; chunks 2–4 refused none (campaign chunks 62–65: 0, 0, 815, 2500).[RJ][CP]

2. What JFNK-1 changed

Measured against chunk 61 (Fig. 1, red curve), the JFNK-1 last iterate differs by up to 0.226 dex in ρ (cell 225, $r = 1.231\,R_p$; 446 cells outside 0.004 dex), 0.075 dex in T (cell 136, $r = 1.048\,R_p$), and 0.087 in the v measure ($r = 1.872\,R_p$; 262 cells above 0.01). The largest change is in the **helium distribution**: the local He/H nucleus ratio, $(n_{\mathrm{HeI}} + n_{\mathrm{HeII}} + n_{\mathrm{HeIII}} + n_{\mathrm{HeH^+}})/n_{\mathrm{H,nuclei}}$ with $n_{\mathrm{H,nuclei}} = n_{\mathrm{HI}} + n_{\mathrm{HII}} + 2n_{\mathrm{H_2}} + 2n_{\mathrm{H_2^+}} + 3n_{\mathrm{H_3^+}} + n_{\mathrm{HeH^+}}$ (He I includes the 2^3S population, which is not added again), is 0.0534 at $1.2\,R_p$, 0.0168 at $2\,R_p$ and 0.0035 at $10\,R_p$ in chunk 61, and 0.0875, 0.0719 and 0.0447 in the JFNK-1 iterate (up to 13 times larger). The H_2 fraction $x_2 = 2n_{\mathrm{H_2}}/(n_{\mathrm{HI}} + n_{\mathrm{HII}} + 2n_{\mathrm{H_2}})$ falls from 0.51–0.56 to 0.33–0.36 between 2 and $10\,R_p$. The cell-centered spread of $\rho v r^2$ over $r \geq 2\,R_p$ falls from 5.56×10^{-2} to 2.30×10^{-4}, and $4\pi\rho v r^2$ at cell $N-20$ from 7.282×10^7 to 7.084×10^7 g s^{-1} (−2.7%).[T:distance, massflux, heh_x2]

The first pass made a substantial element-composition update: its reported element displacement was 0.549, while the carrier displacement was 0.016. These scalar measures (the maximum change of an elemental mass fraction normalized by its reservoir value) do not establish how much of the final He/H profile change occurred in that pass.[T:outer_passes_jfnk1] In chunk 61 the normalized stationary residual of the elemental He/H row is 0.311 at cell 4; this is a balance residual relative to the row's terms, not a 31% error in the helium abundance. The carrier residuals range from 9.3×10^{-5} (H_3^+) to 4.8×10^{-3} (He 2^3S), except H_2 at 0.151 (cell 3). These are much smaller than the carrier residuals during the stationary iterations, but neither the carrier rows nor the elemental row satisfy the 10^{-5} stationary tolerance.[R1 console 120–160]

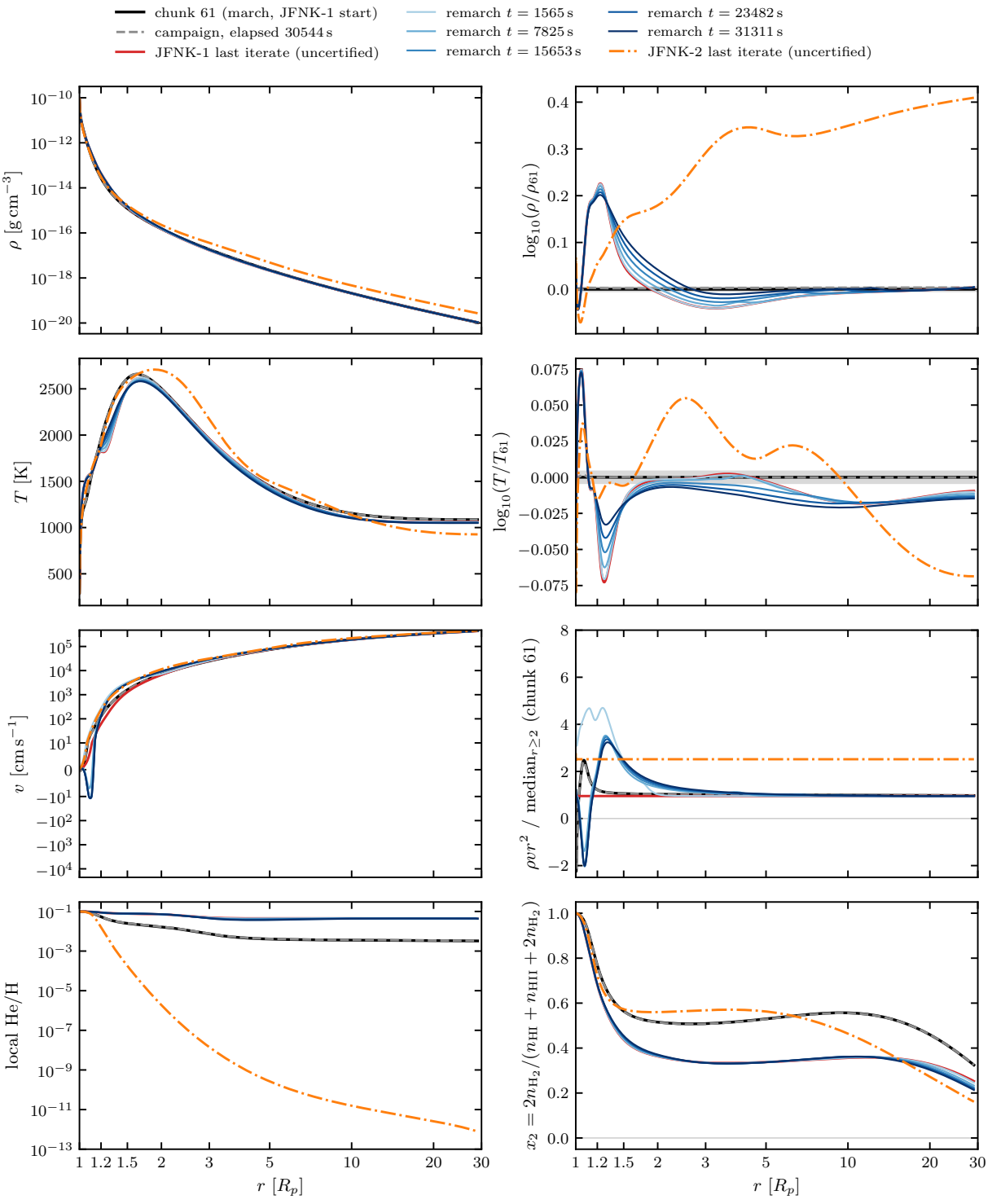


Figure 1: Fluid profiles. Black: chunk 61; gray dashed: campaign at elapsed time 30 544 s; red: JFNK-1 last iterate (uncertified); blue (light to dark): remarch at 1565, 7825, 15 653, 23 482 and 31 311 s; orange dash-dot: JFNK-2 last iterate (uncertified). Right column: differences from chunk 61; the gray band is ± 0.004 dex. The mass flux is normalized by the median of chunk 61 over $r \geq 2R_p$.

3. A. The carriers when the JFNK-1 iterate is marched

Figures 2–4 and Table 2. Two groups appear.

Fast ions near the layer top (1.1–2 R_p). Where the JFNK carrier rows fail (cells 200–240, $r = 1.15$–$1.30\,R_p$), the JFNK-1 iterate carries more H_2^+, H_3^+ and HeH^+ than chunk 61 (up to +0.4 dex at 1.1–1.3 R_p for H_2^+ and H_3^+, +0.9 dex for HeH^+). By the first frame (1564.5 s) these fall by 0.40–1.03 dex at 1.20–1.50 R_p, below the chunk-61 values, while at 1.27–2 R_p He^+ and He(2^3S) rise by 0.16–0.36 dex. The code itself quotes an H_2^+ lifetime of about 10^{-3} s (console, `molecular_reaction_heat`); with the frames 1564.5 s apart, we can only say that this change is present at the first frame, which also includes the load re-equilibration (Section 1.). Afterwards H_2^+ and HeH^+ partly rebound (+0.30 and +0.43 dex at 1.27 R_p, half of that change reached at $\approx 1.5 \times 10^4$ s), apparently following the slower fluid change below.

Slow change in the layer (1.03–1.5 R_p). H^+ and electrons change by at most 0.12 dex in the first frame and then steadily: at 1.06 R_p H^+ falls by 0.75 dex and e^- by 0.49 dex, at 1.27 R_p both rise by 0.37–0.39 dex; at these cells half of the measured change is reached after 6000–8500 s (at 1.50 R_p after 11 500 s); He(2^3S) at 1.03–1.1 R_p falls by a further 0.24–0.47 dex. Over the last chunk (23 482–31 311 s) these quantities still move by 0.01–0.06 dex, so they have not settled. This is where the velocity reverses sign in the remarch (inflow of up to 12 cm s^{-1} at 1.034–1.127 R_p, cells 117–191, at 31 311 s; outflow in chunk 61 and in JFNK-1; Fig. 1).

Wind ($r \geq 3\,R_p$) and neutrals. Changes from the JFNK-1 iterate are at most 0.34 dex (He(2^3S) at 3 R_p; He^+ 0.27 dex there) and below 0.2 dex beyond 5 R_p. H I, He I and H_2 change by at most 0.05, 0.06 and 0.08 dex at any of the sampled cells. The helium excess of the JFNK iterate is carried along: relative to the campaign at equal time, He I and He^+ remain 0.7–1.06 dex higher beyond 3 R_p throughout (Fig. 2).

By the first saved frame, the fast-ion densities have changed strongly within the displaced fluid and helium distribution. They have not returned to the chunk-61 values. By the end of chunk 4 the stored state's carrier rows are 2.1×10^{-3} to 1.8×10^{-2} (H_2 0.109), as measured by JFNK-2 on load (its console lines 130–151): far below the 0.5–0.85 the JFNK passes leave, though about ten times the chunk-61 values (9×10^{-5} to 4.8×10^{-3}).

Table 2: Carrier changes in the remarch at selected cells [`carrier_timescales.txt`]: Δ_1 = change of $\log_{10} n$ from the JFNK-1 iterate to the first frame (1564.5 s), Δ_2 = change from the first frame to 31 311 s, $t_{1/2}$ = first time (interpolated between frames, measured from the remarch origin) at which half of the finite change Δ_2 is reached; it is not a chemical half-life or evidence of eventual equilibrium (for example, the small He(2^3S) change at cell 217 gives 1642 s despite later nonmonotonic changes). Δ_1 includes the load re-equilibration.

cell	r [R_p]	species	Δ_1 [dex]	Δ_2 [dex]	$t_{1/2}$ [s]
149	1.060	H^+	-0.12	-0.75	7597
149	1.060	H_2^+	+0.11	-0.42	6377
149	1.060	HeH^+	+0.22	-0.43	6574
149	1.060	He(2^3S)	-0.48	-0.47	5929
217	1.201	H^+	+0.02	+0.04	4359
217	1.201	H_2^+	-0.45	+0.11	11517
217	1.201	HeH^+	-0.64	+0.16	12603
217	1.201	He(2^3S)	-0.02	+0.00	1642
234	1.270	H^+	+0.08	+0.39	7343
234	1.270	H_2^+	-0.65	+0.30	14995
234	1.270	HeH^+	-1.03	+0.43	15122
234	1.270	He(2^3S)	+0.34	+0.21	5264
269	1.499	H^+	+0.05	+0.40	11484
269	1.499	H_2^+	-0.40	+0.19	14803
269	1.499	HeH^+	-0.61	+0.19	15876
269	1.499	He(2^3S)	+0.36	+0.30	10263
309	2.003	H^+	+0.01	+0.12	17139
309	2.003	H_2^+	-0.18	+0.01	19979
309	2.003	HeH^+	-0.37	+0.01	2690
309	2.003	He(2^3S)	+0.36	+0.12	11367

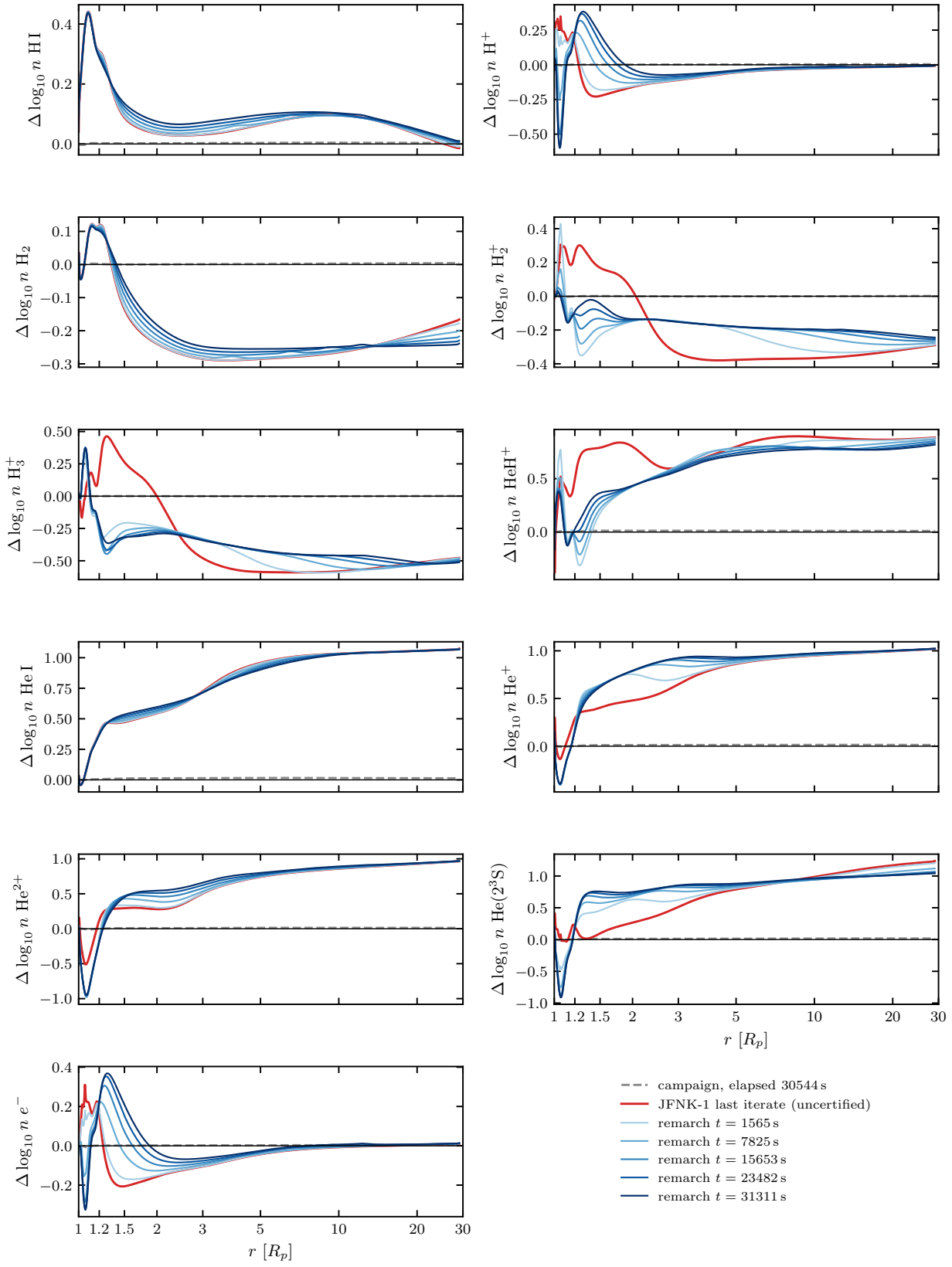


Figure 2: $\log_{10}$ of each density relative to chunk 61. Curves as in Fig. 1, without JFNK-2 (shown in Fig. 6).

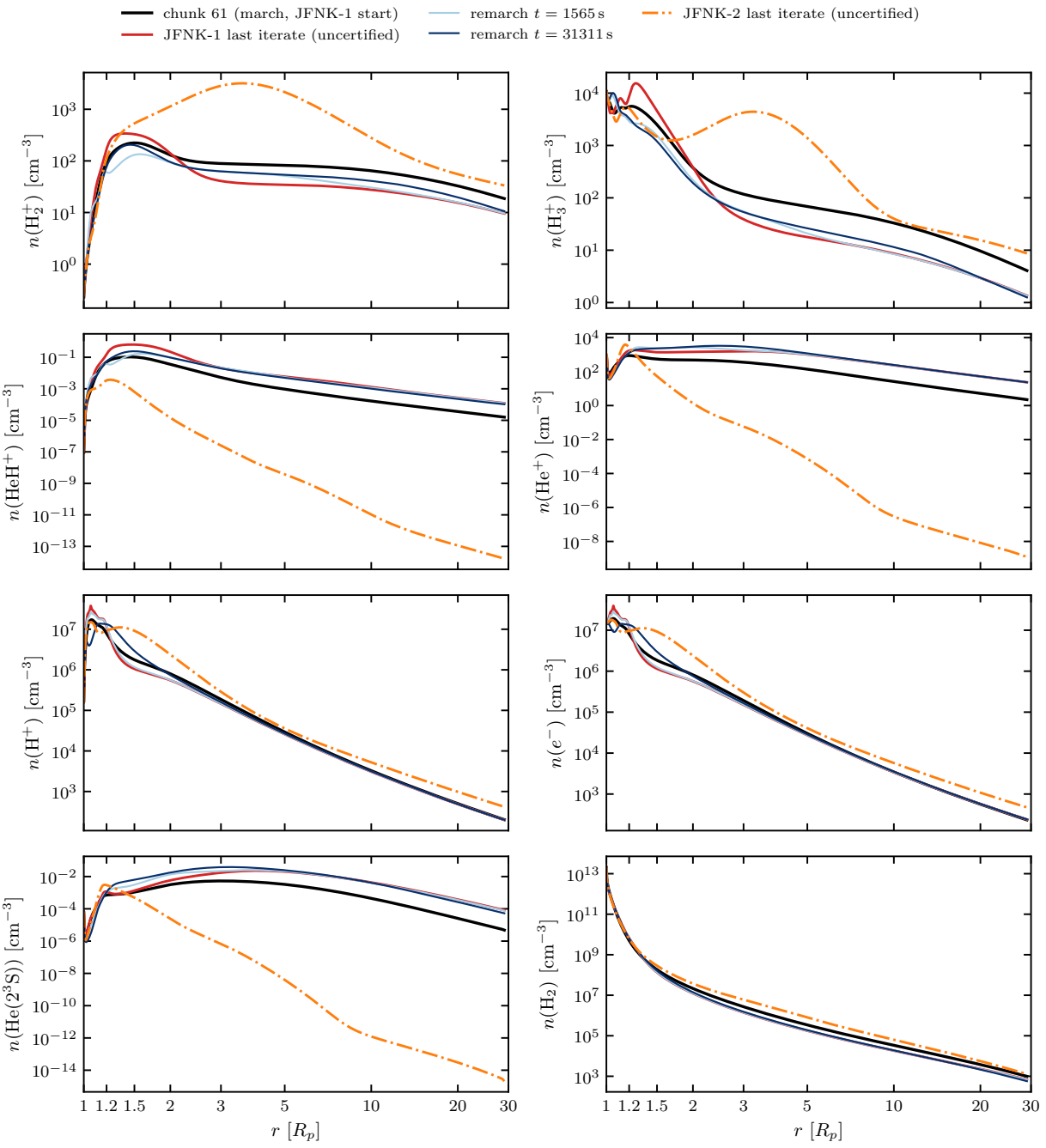


Figure 3: Carrier and H_2 densities: chunk 61, JFNK-1 last iterate, the first and last remarch frames, and the JFNK-2 last iterate.

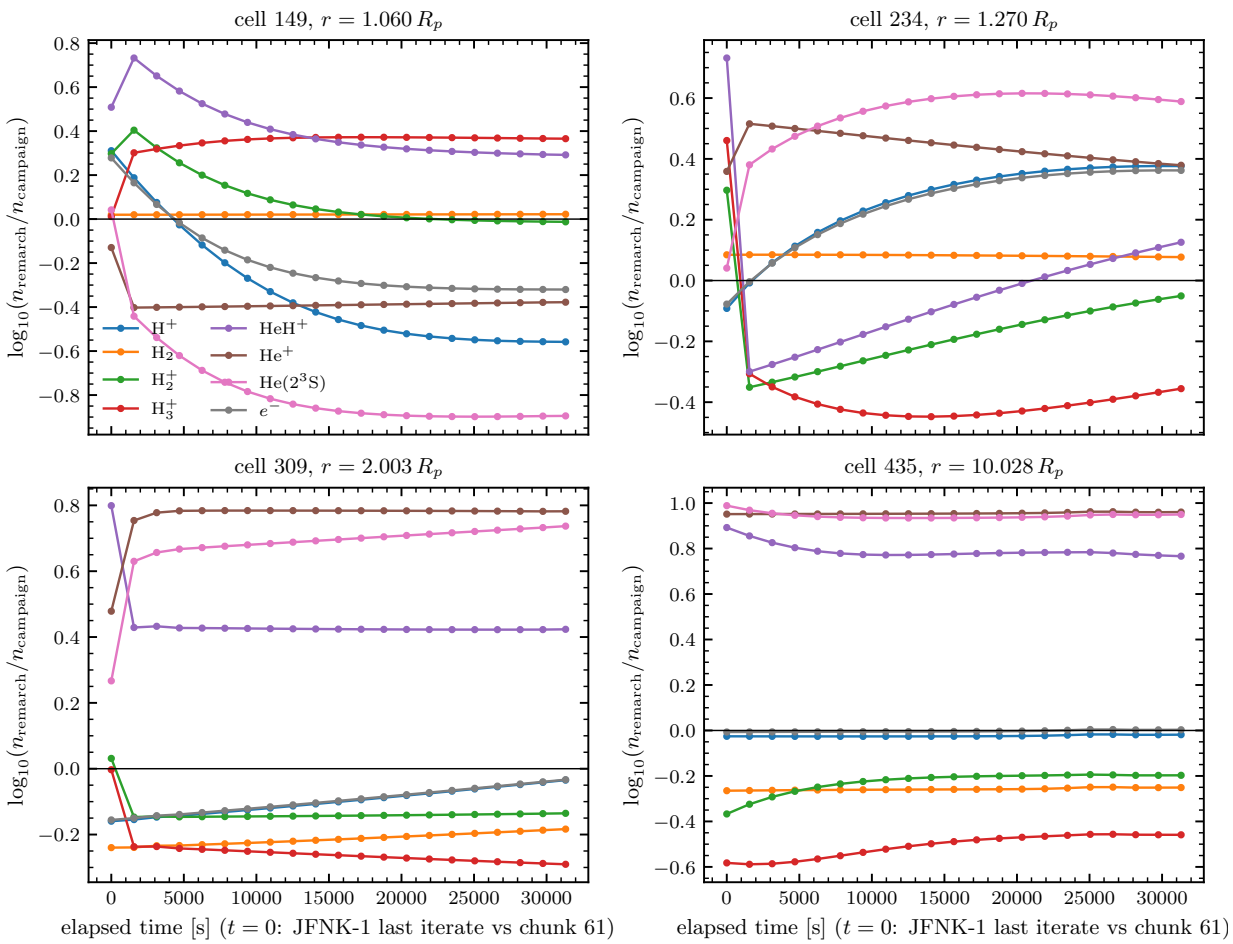


Figure 4: Remarch densities relative to the campaign at equal elapsed time, at four cells. The point at $t = 0$ is the JFNK-1 iterate relative to chunk 61; zero means the remarch has returned to the campaign value.

4. B. The fluid: does it return to the marching state?

There is no overall return within 31 311 s (0.106 $\tau_{\rm cross}$; Fig. 5, Table 3). The maximum density and temperature differences decrease modestly, while the density rms is nearly unchanged and the velocity difference grows. Against the campaign at equal elapsed time, the largest $|\Delta \log_{10} \rho|$ goes from 0.226 (JFNK-1 vs chunk 61) to 0.225 at the first frame and 0.199 at 31 311 s, always at $r = 1.231\,R_p$; its rms over the 500 cells stays at 0.082–0.083. The largest $|\Delta \log_{10} T|$ goes from 0.075 to 0.071 (rms 0.035 to 0.031). The v measure grows from 0.087 to 0.33 at the first frame and 0.42 at 31 311 s; 434–461 cells stay outside the 0.004 dex ρ band and 389–489 outside the T band. Over the same time the campaign moved from chunk 61 by at most 0.0039 dex in ρ, 0.0020 dex in T and 0.0013 in v; the remarch moved away from the JFNK-1 iterate by 0.064 dex in ρ (at $1.54\,R_p$), 0.041 dex in T (at $1.27\,R_p$) and 0.52 in v.

Table 3: Distances over the 500 physical cells [`distance_vs_time.txt`]. v: the 1% rule measure of `marching_acceptance.velocity_band` on the two-state set. Limits: 0.004 dex for T and ρ, 0.01 for v.

pair	t [s]	ρ max (rms) [dex]	T max (rms) [dex]	v max (rms)	cells out $\rho/T/v$
JFNK-1 – chunk 61	0	0.2261 (0.0825)	0.0746 (0.0349)	0.087 (0.029)	446/392/262
remarch – campaign	1565	0.2251 (0.0826)	0.0745 (0.0347)	0.332 (0.093)	448/389/264
remarch – campaign	7825	0.2198 (0.0824)	0.0738 (0.0338)	0.339 (0.105)	452/403/262
remarch – campaign	15653	0.2127 (0.0824)	0.0729 (0.0327)	0.386 (0.112)	461/475/270
remarch – campaign	23482	0.2057 (0.0825)	0.0720 (0.0319)	0.409 (0.116)	459/489/252
remarch – campaign	31311	0.1993 (0.0830)	0.0712 (0.0313)	0.420 (0.117)	434/489/249
remarch – JFNK-1	1565	0.0035 (0.0010)	0.0021 (0.0005)	0.391 (0.111)	0/0/131
remarch – JFNK-1	31311	0.0638 (0.0247)	0.0406 (0.0103)	0.515 (0.129)	274/232/228
campaign – chunk 61	30544	0.0037 (0.0023)	0.0019 (0.0006)	0.001 (0.001)	0/0/0

Where the change happens. In ρ and T the remarch change is gradual: 0.0035 dex in ρ and 0.0021 dex in T at the first frame, growing to 0.064 and 0.041 dex at 31 311 s at an almost constant rate (Fig. 5). In v, 0.39 of the final 0.52 is present at the first frame. The mass-flux spread du over $r \geq 2\,R_p$ rises from 2.3×10^{-4} (the JFNK-1 iterate) to 7.7×10^{-2} at step 10 000, peaks at 0.583 at step 16 920 of chunk 1 (about 2640 s, a time interpolated between saved clocks), and then grows slowly from 0.31 to 0.52. For campaign chunks 62–65, du decreases from 0.05565 to 0.05378. The relative change of the conserved state over one step, $dtu = \max|1 - u/u_{\rm old}|$ (EXHALE_main.f90 3556–3583; not a stationary residual), peaks at 1.0065×10^{-4} (chunk 1, step 12 619) in the raw remarch sequence and reaches 1.2681×10^{-6} at chunk 4 step 50 000 and 6.3406×10^{-7} at step 50 001; campaign values at step 50 000 remain near 4.4–4.5×10^{-8}. The excess mass flux sits at 1.05–2 R_p: at 31 311 s, $\rho v r^2$ is −2.0 to +3.2 times the chunk-61 median at 1.05–1.5 R_p, with inflow at 1.034–1.127 R_p; $4\pi\rho v r^2$ at cell $N-20$ stays at 7.08×10^7 g s^{-1} (campaign 7.35×10^7). A density excess travels outward from 1.5 to about 3 R_p during the four chunks (Fig. 1).

Time scales. From the states we evaluate $\tau_{\rm cross} = 2.91$–3.35×10^5 s (flow crossing beyond Mach 0.01; `marching_acceptance.crossing_times`) and a sound crossing of the whole domain $\sum \Delta r/(c_s + |v|) = 5.84 \times 10^4$ s; 31 311 s is 0.106 $\tau_{\rm cross}$ and 0.54 sound-crossing times. The advection time $\sum \Delta r/|v|$ is 2.3×10^6 s over 1.1–1.2 R_p, 8.3×10^5 s over 1.2–1.5 R_p, 1.7×10^5 s over 1.5–2 R_p and 2.6×10^5 s over 2–30 R_p in chunk 61 [`segment_times.txt`].

A difference that does not decay within the march is the helium distribution of the layer below $2\,R_p$ (He/H at 1.2–2 R_p changes by at most 0.0011, 1.5%, in the remarch, and in the direction away from chunk 61). The campaign itself drifts toward more helium: its He/H at $1.2\,R_p$ rises from 0.05295 to 0.05384 between elapsed −22 554 and +25 747 s (0.0072 dex), at $10\,R_p$ by 0.018 dex. If these slopes of $\log_{10}$(He/H) were held constant, the campaign would reach the JFNK-1 ratios at 1.2 and $10\,R_p$ after about 1.4×10^6 and 3.0×10^6 s, or 5 and 10 chunk-61 crossing times; a constant slope of the ratio itself would instead give 1.8×10^6 and 1.4×10^7 s. These are illustrative extrapolations of a short record, not predictions, lower bounds of the return time of the remarch, or acceptance conditions. The rms ρ distance of the remarch shows no decrease. The remarch does not return to the campaign state within the measured 0.106 crossing times; the different helium distribution and the long advection times are consistent with persistence of the disturbance, but these runs do not establish its cause or its eventual relaxation time. These stationary diagnostics and the helium drift do not change the campaign's marching acceptance criterion (the T, ρ and v bands over the final $K = 3$ crossing-time window); they add no required waiting time.

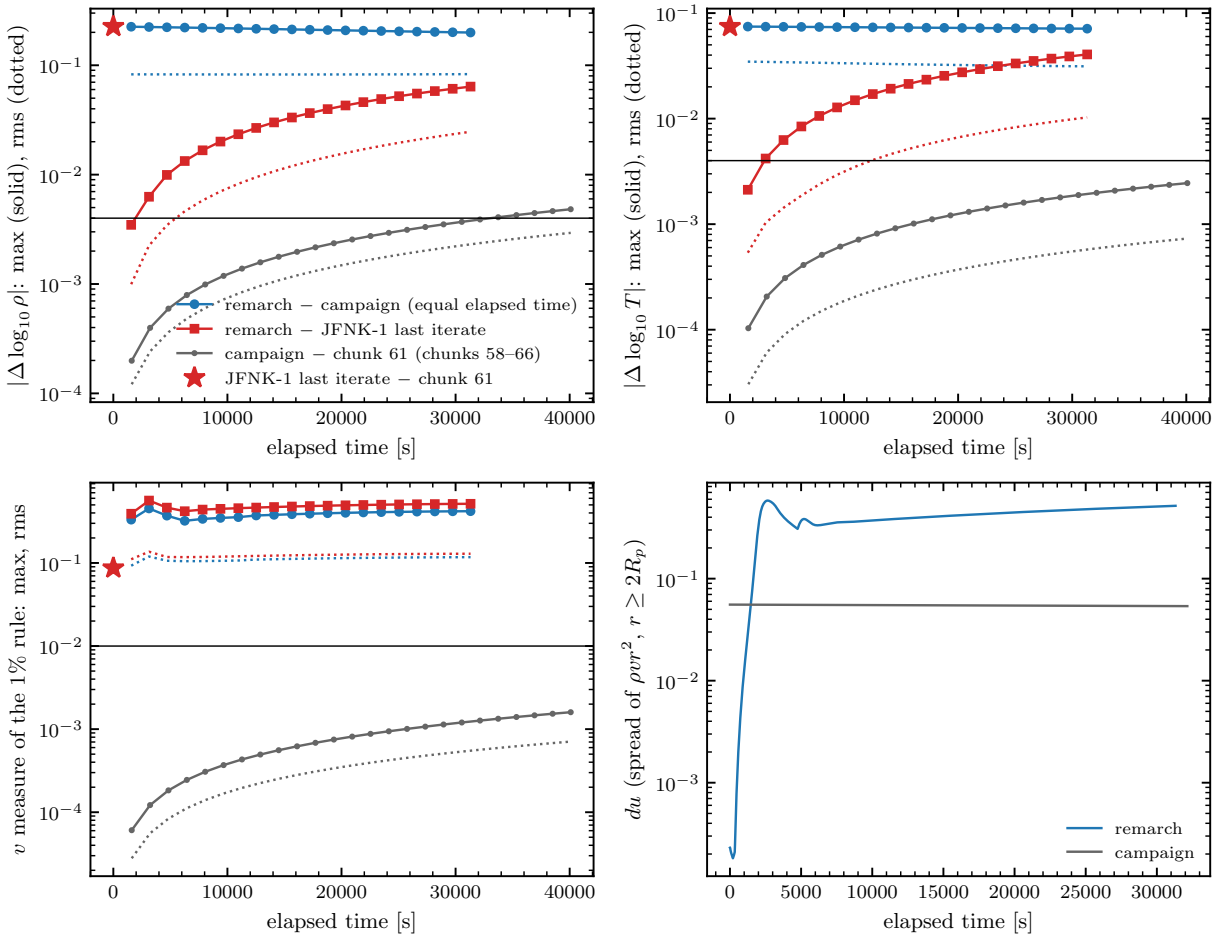


Figure 5: Distances in time. Blue: remarch against the campaign at equal elapsed time; red: remarch against the JFNK-1 iterate; gray: campaign against chunk 61; star: JFNK-1 iterate against chunk 61. Solid: maximum over cells; dotted: rms. Horizontal lines: the 1% limits. Lower right: du from the console records of every step (remarch, campaign chunks 62–65).

5. C. JFNK-2 compared with JFNK-1

Pass histories (Fig. 7; `outer_passes_jfnk{1,2}.txt`). Both attempts used all 60 passes with `info=1`. Hydrodynamic `info=0` in every pass except 6, 49, 51 (JFNK-1) and 7 (JFNK-2). In these passes, `info=2` caused the failed hydrodynamic iterate to be discarded and the last converged wind and its composition to be restored (EXHALE_main.f90 8148–8164); the composition update was then recomputed with $\omega = 0.5$ and a reduced carrier movement bound (8660–8669), and the next hydrodynamic solve occurred in the following outer pass. Wall time per pass over all 60 passes: median 263.975 and 264.125 s, mean 331.849 and 268.903 s; JFNK-1 had six passes above 700 s (passes 5, 6, 7, 56, 57, 59), JFNK-2 none above 400 s. JFNK-2 ran on lart3 with load average 23.4 at its start (JFNK-1: 12.1).

The two starts lead to similar late-pass diagnostics. JFNK-2 started from a state whose carrier rows were 2.1×10^{-3}–1.8×10^{-2} (JFNK-1: 9×10^{-5}–4.8×10^{-3}) and whose He/H already had the JFNK-1 distribution. Its first worst row was 0.71 (JFNK-1: 0.40), and it had no approach phase: 0.48–0.64 in passes 2–8 against 0.08–0.17 for JFNK-1 in passes 2–6. At passes 28–35 the worst rows of the two attempts differ by up to 0.065 and the worst cells by up to 59. From pass 36 to pass 48 the records are similar: the same worst species, cells within two of each other, worst rows within 0.03, similar element and carrier displacements (e.g. pass 37 carrier displacement 0.333 and 0.353), and the same kind of ending of the carrier update (e.g. pass 40: He(2^3S) at cell 500, 0.998 in both; pass 45: H_2^+ at cell 343, 0.836 and 0.835). JFNK-2 passes 49–59 resemble its passes 32–42 in the printed diagnostics (largest difference of the worst row 0.007, same species); this is not an exact repeat of all entries (pass 35 against 52: worst cell 365 against 309) or a comparison of full states. JFNK-1 departs from the similar diagnostic sequence at pass 49 (`info=2`).

Where the JFNK-2 last iterate lies (Fig. 6; `jfnk2_compare.txt`; maxima over cells, rms in brackets).

pair	ρ [dex]	T [dex]	v measure
JFNK-2 – its start (remarch end)	0.406 (0.219) at $28.5\,R_p$	0.062	0.23
JFNK-2 – JFNK-1	0.405 (0.231) at $28.5\,R_p$	0.066	0.79
JFNK-2 – chunk 61	0.409 (0.216) at $29.0\,R_p$	0.081	0.65
JFNK-1 – chunk 61	0.226 (0.082) at $1.23\,R_p$	0.075	0.09

JFNK-2 removes helium from the wind: He/H is 0.0154 at $1.2\,R_p$, 1.8×10^{-4} at $1.5\,R_p$, 1.8×10^{-6} at $2\,R_p$ and 1.5×10^{-11} at $10\,R_p$, against 0.0875, 0.078, 0.072 and 0.045 in JFNK-1 and 0.053, 0.025, 0.017 and 0.0035 in chunk 61. Its x_2 at 2–5 R_p (0.56–0.57) is close to chunk 61 (0.51–0.53), unlike JFNK-1 (0.34–0.36). Its density is 0.18–0.29 dex higher than chunk 61 at 2–3 R_p and 0.29–0.41 dex higher beyond $3\,R_p$, and its $4\pi\rho v r^2$ at cell $N-20$ is 1.869×10^8 g s^{-1}, 2.6 times chunk 61 (7.28×10^7) and JFNK-1 (7.08×10^7), with spread $du = 1.8 \times 10^{-4}$.

Is the displacement reproducible? Not as a state. Over the 500 cells, the JFNK-2 displacement (JFNK-2 minus its start) is *anti*-correlated with the JFNK-1 displacement (JFNK-1 minus chunk 61): Pearson coefficient −0.60 in $\log\rho$, −0.48 in $\log T$, −0.99 in log He/H. The march between them returned almost nothing of the JFNK-1 displacement: the least-squares slope of the march change on the JFNK-1 displacement is −0.024 in $\log\rho$ and −0.13 in $\log T$. What recurs is a similar late-pass diagnostic record and a large helium contrast between the two saved pass states of each attempt (Section 6.): the JFNK-1 last iterate is helium-rich, the JFNK-2 last iterate helium-poor. The kept states of the same phase from the two attempts are closer to each other than to the opposite phase: the two helium-rich states (JFNK-1 `_p59`, JFNK-2 `_p58`) differ by 0.22 dex in ρ, 0.068 dex in T and 0.16 dex in He/H, while consecutive states

of one attempt differ by 0.89 (JFNK-1) and 1.66 dex (JFNK-2) in ρ. The saved states and scalar diagnostics appear consistent with a strongly varying partitioned iteration that does not approach the marching state. They do not establish a common periodic orbit or an alternative stationary solution.

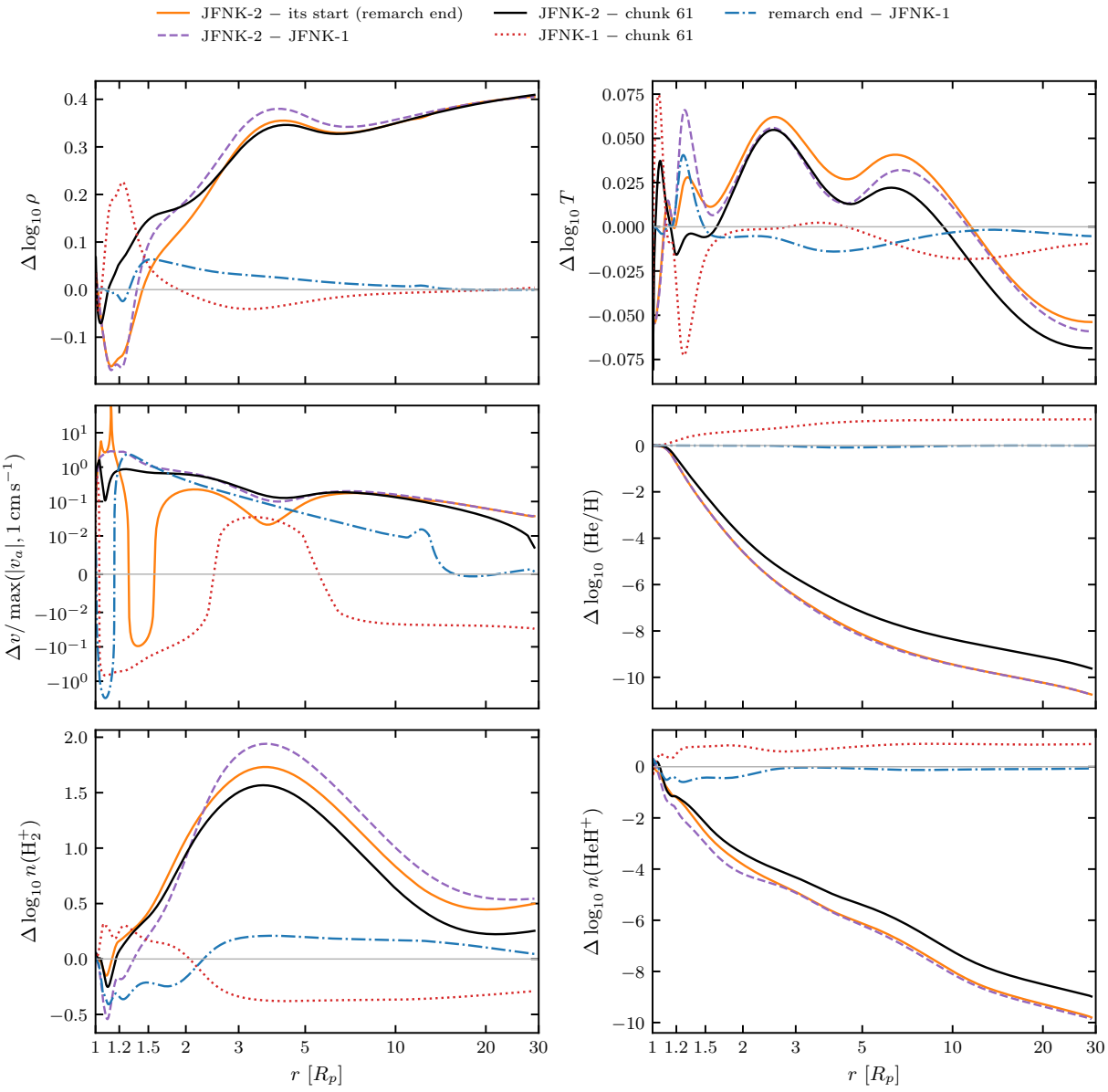


Figure 6: Cellwise differences between the JFNK-2 last iterate, its start (the end of the remarch), the JFNK-1 last iterate and chunk 61. The velocity panel shows $\Delta v / \max(|v_a|, 1\,\mathrm{cm\,s^{-1}})$, a the reference state of each pair, on a symmetric logarithmic scale.

6. D. How the state moves during the outer passes

What is kept. Each attempt publishes the state after the composition update of a pass into `output/pass_state/` and keeps only the last two: JFNK-1 holds `_p58` and `_p59` (both complete, with manifest, Hydro, Ion and conserved

files; `current` names `_p59`, which is not empty when read on 2026-11-12), and JFNK-2 holds `_p58` and `_p59`. Earlier pass states are not available, so the motion over the passes is read from the console lines of each pass (Table files `outer_passes_jfnk{1,2}.txt`, Fig. 7).

JFNK-1, passes 1–8: an approach. The first element relaxation reports a displacement of 0.549; the following element displacements fall to 0.035–0.09 (passes 4–8), the carrier updates are limited by the trust radius, which doubles from 0.01 to 0.16, and the worst carrier row after the hydrodynamic solve falls from 0.40 to 0.08 (passes 5–6; H_2^+ at cell 243, $1.32\,R_p$).

Passes 9–60: no further approach; a recurring pattern of pass types. At pass 9 the element relaxation stops on its step budget with a displacement of 0.87, and from then on the worst row stays at 0.42–1.0. From pass 19 a pattern recurs every 2–4 passes: a pass whose carrier relaxation ends because the shortest admissible trial interval was not covered, with a carrier displacement of 2.5–2.7 and the worst carrier residual in He^{2+} at 8–29 R_p; then a pass with composition under-relaxation $\omega = 0.5$ ending on the movement bound (displacement 0.97); then one or two passes ending on the step budget, whose returned carrier residual is 5×10^{-5}–5×10^{-4}. In passes 38–46 the element relaxation applies 0.88–0.91 in every pass, alternating between the two kinds of carrier update. The worst row sits in two groups of cells: 200–243 (1.15–1.32 R_p, mostly H_2^+, HeH^+) and 278–357 (1.58–3.32 R_p, mostly He^+, H_2^+), with $He(2^3S)$ at the outer boundary (cells 426–500) in passes 40, 42, 44. The hydrodynamic solve returned `info=2` in passes 6, 49 and 51; each failed iterate was discarded and the composition update was recomputed from the restored converged wind with $\omega = 0.5$. It returned `info=0` otherwise. The H_2 "front" printed by the code (innermost cell with $x_2 < 0.5$) jumps between 1.23 and 22.6 R_p because x_2 stays near 0.5 over most of the wind; it is not a useful position measure here.

The size of the change between passes. The two kept JFNK-1 states differ enormously: wind He/H is 3.8–5×10^{-8} beyond $10\,R_p$ in `_p58` and 0.044–0.045 in `_p59`; ρ differs by up to 0.89 dex (4.2 R_p), T by 0.17 dex and the v measure by 0.47. The step from `_p59` to the final iterate (the hydrodynamic solve of pass 60 with its usual thermochemical and boundary refreshes, no element or carrier update; He/H changes by less than 2×10^{-14} dex) moves ρ by up to 0.23 dex, T by 0.13 dex and v by 0.33 (Fig. 8). The final iterate of an attempt thus appears to be one state of a strongly varying sequence, not the end of a drift. Two adjacent saved states cannot establish a sustained alternation of full profiles, a periodic orbit, or independence from the initial state; they show that adjacent passes moved the state by more than the JFNK-1 to chunk-61 difference. In JFNK-2 the two kept states are the other way round: `_p58` is helium-rich (He/H 0.064 at $10\,R_p$) and `_p59` helium-poor (1.5×10^{-11}); they differ by 1.66 dex in ρ and 0.27 dex in T, and the hydrodynamic solve of pass 60 moves ρ by 1.62 dex. The mass-loss measure $4\pi\rho v r^2$ at cell $N-20$ of the kept states is 2.1×10^7 and 1.22×10^8 g s^{-1} (JFNK-1 `_p58`, `_p59`) and 2.07×10^8 and 8.5×10^6 g s^{-1} (JFNK-2), i.e. it changes by a factor 6–24 from one pass to the next.

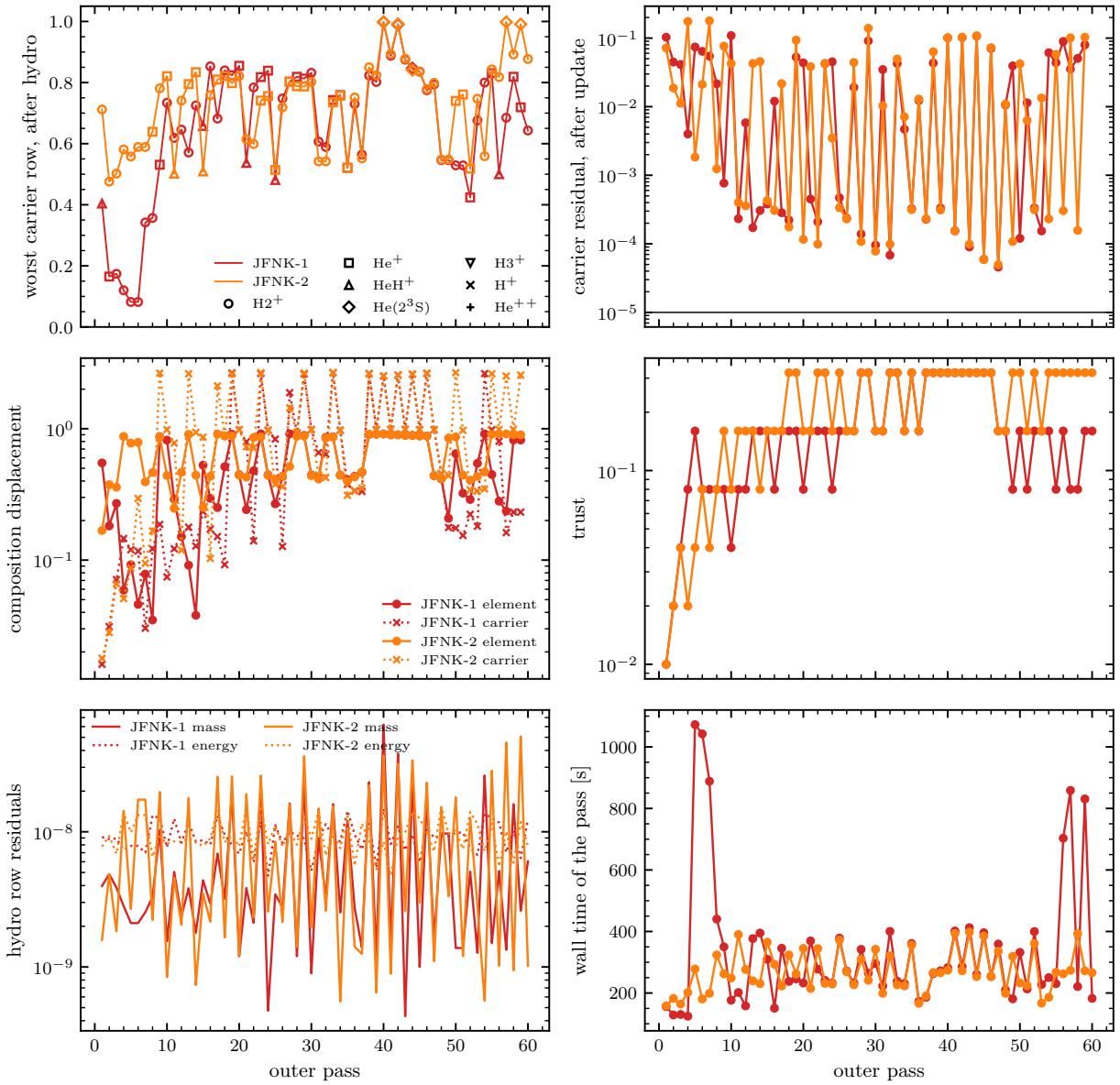


Figure 7: Outer-pass histories of JFNK-1 (red) and JFNK-2 (orange). Upper left: worst gated carrier row after the hydrodynamic solve (transported fractions retained, with the usual thermochemical and boundary refreshes), with the species; upper right: carrier residual of the composition returned by the update of the pass (horizontal line: 10^{-5}); middle: element and carrier displacements and the trust radius; lower: hydrodynamic row residuals and wall time of each pass.

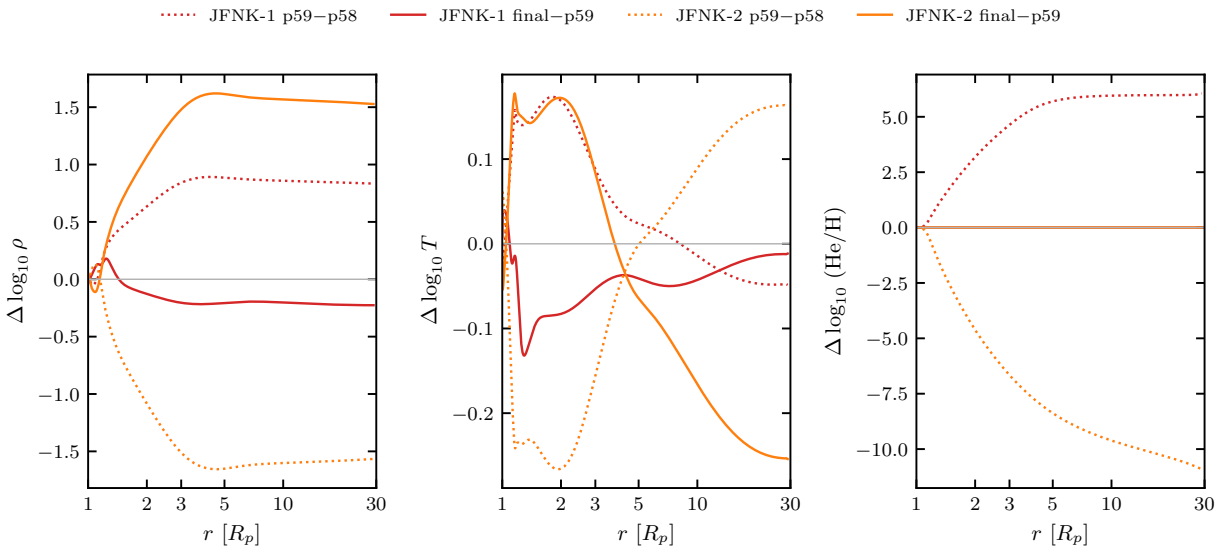


Figure 8: Motion between the saved intermediate iterates. Dotted: `_p59` minus `_p58` (one hydrodynamic solve and one composition update). Solid: the uncertified last iterate minus `_p59` (the final hydrodynamic solve with its usual thermochemical and boundary refreshes, without a subsequent element or carrier update).

7. E. Why the carrier rows do not close: a cautious reading

What the row is. The carrier balance entry of the certification is, for each transported carrier and cell, the stationary form of its transport equation: the divergence of its diffusive, eddy and drift face fluxes, plus the divergence of its advective face flux, minus the net chemical source, with the time term removed ($\Delta t = 10^{30}$) (`carrier_steady_residual` and `carrier_residual`, `diffusive_photochemistry.f90` 4288–4525, 7402–7535). It is normalized by the sum of the absolute diffusive face contributions, the advective magnitude, the gross chemical production and loss, and an absolute floor (7417–7425, 7453, 7503, 4491). The floor is 10^{-20} times a carrier-specific elemental reference density (free hydrogen for H_2, H^+, H_2^+, H_3^+; total helium nuclei for the helium stages and $He(2^3S)$; the smaller of the two for HeH^+; 5771–5804) times the cell's signal-crossing rate. The tolerance is 10^{-5}. A measure $q = 0.5$–0.85 represents an imbalance of 50–85% of that sum. Only when transport and the floor are negligible is $q = |P-L|/(P+L)$, implying $\max(P,L)/\min(P,L) = (1+q)/(1-q)$, about 3–12.3. The corresponding volume-weighted measures are also large; pass 30 of JFNK-1 gives 0.54, 0.33 and 0.44 for He^+, H_2^+ and HeH^+ [`handed_back_rows_jfnk1.txt`]. These integrated measures do not by themselves determine the number or radial extent of the affected cells.

What the records show.

1. In marching states these rows are much smaller, though still above 10^{-5}: 9.3×10^{-5}–4.8×10^{-3} in chunk 61 (H_2 0.15 at cell 3), 2.1×10^{-3}–1.8×10^{-2} at the end of the remarch (H_2 0.11).

2. Inside the JFNK passes, the composition updates reduce the maximum carrier residual to 4.6×10^{-5}–3.3×10^{-4} in the listed step-budget-limited passes, while remaining above the 10^{-5} stationary tolerance (JFNK-1 passes 30, 32, 39, 41, 43, 45, 47; Fig. 7).

3. The next hydrodynamic solve, with the transported fractions retained (console line 29: "hydrodynamic sweeps

retain transported x(H II), x(He II) and x(He III)") and the usual thermochemical and boundary refreshes applied (EXHALE_main.f90 8043–8080), raises the worst gated carrier residual to 0.55–1.0 in the listed passes (31, 33, 40, 42, 44, 46, 48 of JFNK-1). The transition from the saved pass-59 state to the uncertified last iterate contains this hydrodynamic solve and its refreshes, with no subsequent element or carrier relaxation; in it T changes by up to 0.13 dex ($1.30\,R_p$), ρ by up to 0.23 dex, and v at $1.26\,R_p$ by −61% in JFNK-1; in JFNK-2 the same step changes T by up to 0.25 dex and ρ by up to 1.62 dex.

4. The helium partition does not settle: the element relaxation applies a displacement of 0.88–0.91 in every pass from 38 to 46, and the two published pass states of JFNK-1 have wind He/H of 4×10^{-8} (pass 58) and 0.045 (pass 59) (Section 6.).

Reading (tentative). The records appear consistent with a partitioned iteration that does not contract: a hydrodynamic solve with retained transported fractions changes T and ρ by 0.1 dex or more (the two solves that can be isolated, pass 60 of each attempt), which might be enough to unbalance the fast ions by an $O(1)$ fraction of their gross rates; each composition update then moves helium and the carriers by amounts that change the next hydrodynamic solution by a similar amount. This is not a term-by-term causal proof. The march instead advances the coupled time-dependent model with its existing operator splitting and small physical steps of about 0.16 s, far shorter than the $\sim 10^4$ s on which the layer evolves, and the fast-carrier rows stay much smaller (though above 10^{-5}). We have not decomposed the rows into their terms (the row-term record `EXHALE_CARRIER_ROW_TERMS` was off), so which term (production, loss or transport) carries the imbalance in the named cells is not measured. Two consequences can be stated from the measurements alone: (i) with the present route and settings, a JFNK last iterate of this molecular case is not a stationary state of the carrier and helium equations and lies outside the 1% band of the marching state; (ii) marching from such an iterate did not return to the marching state within $0.1\,\tau_{\rm cross}$; the different helium distribution in a layer whose advection times exceed 10^6 s is consistent with that persistence, but the runs do not establish its cause. Using a JFNK iterate as a start for further marching would therefore introduce a disturbance of unknown relaxation time. The campaign state itself is not stationary in He/H either (normalized elemental residual 0.31 at cell 4; He/H rising slowly, Section 4.); which helium distribution the stationary wind has is not decided by these runs.

Not checked

- The composition change made by the load re-equilibration of remarch chunk 1, species by species: no state is written before step 10 001.
- Carrier changes faster than the frame interval of 1564.5 s; the fast-ion time scale is the code's own quoted H_2^+ lifetime, not a measurement here.
- The individual terms (production, loss, transport) of the unbalanced carrier rows; the row-term record was off in all runs.
- The states of JFNK passes other than the last two of each attempt.

- Whether the remarch would return to the campaign state over one or more crossing times; only $0.106\,\tau_{\rm cross}$ was marched, and the statements on longer times are illustrative extrapolations of $\log_{10}$(He/H), not predictions or bounds.
- Which helium distribution (the marching one or the JFNK one) the stationary wind has; neither state is certified and the normalized elemental He/H residual of the marching state is 0.31.
- The physical acceptability of the remarch inflow at 1.03–1.13 R_p and of the pass-to-pass variation of the JFNK iterates; no resolution, CFL or step-size test was made.
- Bitwise or host effects: the campaign ran on lart1, the remarch and both JFNK attempts on lart3; no replay was made.
- The decomposition of the remaining 0.2 dex density difference at $1.23\,R_p$ into composition (mean molecular mass) and structure.
- Mdot from `EXHALE_resolved.out` of each state; we quote $4\pi\rho v r^2$ at cell $N-20$ computed from the profiles.

Sources

Paths relative to `EXHALE_v1.10/`; E = `scratch_jfnk_from_marching_20261111/heh0p1_s2p53/`, P = `LHS1140b/models/marching_campaign_he01_series_20261106/cases/heh0p1_s2p53/phase/`.

- R1: E`run/` (`console.log`, `VERDICT.txt`, `EXIT.txt`, `output/`); PLAN: E`PLAN.txt`; remarch plan E`remarch/PLAN.txt`.
- RJ: E`remarch/journal.txt`, `progress.txt`, `chunks/c000k/console.log`.
- R2: E`remarch/jfnk2/run/`; CP: P`progress.txt`, P`journal.txt`, P`snapshots/`, P`chunks/c006k.log.gz`.
- T: tables and scripts in E`remarch/analysis/` (`ana_common.py`, `march_analysis.py`, `du_series.py`, `carrier_timescales.py`, `outer_passes.py`, `jfnk2_compare.py`, `figures.py`); readers `examples/exhale_io.py` and `LHS1140b/tools/marching_acceptance.py`.
- Source code: `src/EXHALE_main.f90` (step record 4194, `mass_flux_spread` 4968–4995, H_2 front 9380–9420); `src/modules/lower_atmosphere/diffusive_photochemistry.f90` (`carrier_steady_residual` 4288–4525, `carrier_residual` 7170–7565); `src/modules/time_step/certification.f90` (carrier entries 1898–2200).